MA2760: Mathematical Investigations in Python

Coursework I Specification Summary (40% Weighting) | Module Leader: Rhyd Lewis

1. Background & Core Objective

- **Theoretical Context:** Simulates the 'infinite monkey theorem' to demonstrate how cumulative evolutionary selection differs from pure chance (replicating Richard Dawkins' Weasel experiment).
- **Target Text:** "he who seeks rest finds boredom. he who seeks work finds rest." (62 characters).
- **Search Space:** 28-character alphabet (lowercase a-z, spaces, full stops).

2. Task Breakdown

Task 1: Pure Random Search (Approx. 20%)

- **genRandText Function:** Create a function returning a randomly generated 62-character string over the 28-character alphabet.
- **score Function:** Create a scoring function counting positional character matches against the target quote.
- **Execution Loop:** Repeatedly generate and score strings until the exact target is found or an iteration cap is reached.
- **Statistical Analysis:** Evaluate performance metrics across runs (best, worst, average scores, and score distribution charts).

Task 2: Simulated Evolution (Approx. 40%)

- **Initialization:** Generate an initial candidate string S (length n=62) using genRandText and evaluate its initial score.
- **Mutation Step:** Create offspring S' by cloning S and mutating exactly x=1 character at a randomly selected position to a random character.
- **Selection Criteria:** If score(S') > score(S), replace S with S'. Repeat this cumulative selection loop until target is produced or iteration cap is reached.
- **Comparative Evaluation:** Compare convergence speed and efficiency against Task 1's pure random search.

Task 3: Open-Ended Exploration (Approx. 40%)

- **Mutation Rate Sensitivity:** Investigate whether varying the mutation rate (x) speeds up or slows down convergence to the target text and explain findings.
- **Target Length Sensitivity:** Test algorithm behavior on shorter strings, such as Richard Dawkins' original quote 'methinks it is like a weasel'.
- **Experimental Rigor:** Formulate custom research hypotheses, conduct multiple randomized trials, and perform formal statistical tests to validate conclusions.

3. Code Standards & Report Formatting

- **Code Excellence:** Write neat, elegant, and well-commented Python code. Provide interactive user configuration options (iteration limits, mutation rates) at program startup for extra marks.
- **Visualizations:** Generate all charts using matplotlib.pyplot. Report body should focus on methodologies, experimental logic, and findings rather than line-by-line syntax.
- **Report Formatting:** Typeset using LaTeX template. Aim: 5 pages (max. 8 pages).

4. Submission & Compliance Checklist

- **Deadline:** 12:00pm, Friday, 27th November
- **Hardcopy Submission (School Office):** Printed report, Python code printout, signed Peer Assessment form.
- **Digital Submission (Email to Lecturer):** One email per group with a single zip file named Group6.zip containing Group6.py and Group6.pdf. Do NOT send loose .py files.
- **In person:** Scheduled lab session to explain code logic and demonstrate live execution to Rhyd.
- **Academic Integrity:** Zero tolerance for plagiarism, inter-group code copying, or AI-generated text/code (results in zero marks).